



Cloud Computing

Assignment-Week 7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which of the following applications is least suitable for computation offloading to a remote cloud in Mobile Cloud Computing (MCC)?

- A) Image rendering requiring intensive GPU computation
- B) Large-scale machine learning inference
- C) Emergency collision avoidance requiring millisecond response
- D) Video transcoding

Correct Answer: C

Detailed Solution: Safety-critical real-time applications require extremely low latency. Network delays and connectivity uncertainty make cloud execution unsuitable. Such tasks are better handled locally or at nearby edge/fog nodes.

QUESTION 2:

A cloud-based mobile application automatically allocates additional computing resources during a sudden surge in users and releases them when demand decreases. Which cloud characteristic is demonstrated?

- A) Resource pooling
- B) Rapid elasticity
- C) Location independence
- D) Broad network access

Correct Answer: B

Detailed Solution: Rapid elasticity enables cloud resources to scale up or down automatically according to workload, ensuring efficient resource utilization and performance.



QUESTION 3:

Dynamic computation offloading primarily aims to optimize which combination of objectives?

- A) Screen brightness and storage capacity
- B) Execution time, energy consumption, and resource utilization
- C) Camera quality and display refresh rate
- D) Cloud subscription cost only

Correct Answer: B

Detailed Solution: Dynamic offloading balances performance and energy by considering execution delay, battery consumption, CPU utilization, and network conditions.

QUESTION 4:

A mobile augmented reality application requires object recognition with an end-to-end response time below **20 ms**. Which infrastructure is most appropriate?

- A) A geographically distant public cloud data center
- B) A nearby cloudlet connected through a high-speed local network
- C) Local storage only
- D) An offline mobile application

Correct Answer: B

Detailed Solution: Augmented reality applications are highly latency-sensitive. A nearby cloudlet minimizes communication delay and provides computational resources close to the user, unlike a remote cloud.

QUESTION 5:

A healthcare application stores patient records in the cloud. Which security requirement is most critical to ensure that unauthorized users cannot access sensitive medical information?

- A) Elasticity
- B) Data confidentiality
- C) Resource pooling
- D) Load balancing

Correct Answer: B



Detailed Solution: Medical records contain sensitive information. Confidentiality ensures that only authorized individuals can access the data.

QUESTION 6:

Which limitation of traditional cloud computing is primarily addressed by fog computing?

- A) Limited virtualization support
- B) High network latency caused by distant cloud data centers
- C) Lack of cloud storage
- D) Insufficient computing power in cloud servers

Correct Answer: B

Detailed Solution: Fog computing places computational resources closer to end devices, minimizing the delay associated with transmitting data to distant cloud data centers.

QUESTION 7:

Which statement about fog nodes is **FALSE**?

- A) They may perform filtering and aggregation of sensor data.
- B) They are usually located close to IoT devices.
- C) They permanently store all enterprise data.
- D) They reduce communication with distant cloud servers.

Correct Answer: C

Detailed Solution: Fog nodes mainly perform temporary processing and short-term storage. Long-term storage is generally handled by cloud data centers.

QUESTION 8:

Statement 1: Geospatial Cloud Computing enables parallel processing of massive satellite imagery by distributing computation across multiple cloud nodes.

Statement 2: Distributed storage and parallel processing frameworks reduce execution time for large geospatial datasets compared with a single desktop workstation.



- A) Both Statement 1 and Statement 2 are true, and Statement 2 is the correct explanation of Statement 1
- B) Both Statement 1 and Statement 2 are true, but Statement 2 is not the correct explanation of Statement 1
- C) Statement 1 is true, but Statement 2 is false
- D) Statement 1 is false, but Statement 2 is true

Correct Answer: A

Detailed Solution: Distributed storage and parallel computation are the fundamental reasons cloud platforms efficiently process massive geospatial datasets.

QUESTION 9:

A cloud platform divides a large raster image into tiles for parallel processing. Which challenge must be carefully addressed to ensure accurate results?

- A) Screen resolution of client devices
- B) Boundary effects between adjacent image tiles
- C) Keyboard input latency
- D) Number of GIS users

Correct Answer: B

Detailed Solution: Many raster operations (e.g., filtering, convolution, terrain analysis) require neighboring pixels. Partitioning into tiles can introduce boundary effects unless overlap or edge handling is implemented.

QUESTION 10:

A wearable ECG sensor continuously streams a patient's heart signals. Which architecture is most appropriate to detect cardiac abnormalities within a few milliseconds?

- A) IoT sensors → Remote Cloud → Physician
- B) IoT sensors → Fog/Edge Node → Cloud
- C) IoT sensors → Local Storage → Weekly Upload
- D) IoT sensors → FTP Server → Desktop GIS

Correct Answer: B



NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur



Detailed Solution: Delay-sensitive healthcare applications require immediate processing at nearby fog/edge nodes. The cloud is mainly used for long-term storage, historical analysis, and model training.